

IBM BOB 2.0 HACKATHON

LegacyLift

Safe, verified legacy code modernization

23 → 0

deprecated usages

31

characterization tests

0 → 29

tests passing on Py 3.12

github.com/Abrik-Ai/LegacyLift • abrik-ai.github.io/LegacyLift

The problem

Legacy code is everywhere, and nobody wants to touch it

- No tests, no docs, original authors gone
- Deprecated APIs and outdated runtimes (Python 2, old libraries)
- Upgrades are risky: nobody remembers the exact behavior
- Worst bugs are silent: code still runs, but behaves differently
- Result: modernization is postponed and technical debt grows

Our solution: protect first, then change

A 5-stage pipeline where IBM Bob does the engineering

1 Analyze

Find every deprecated construct, rate risk



2 Protect

Characterization tests, golden outputs from the original runtime



3 Modernize

Upgrade code with subagents, per official guides



4 Verify

Same tests on new code; fix or roll back



5 Report

Before/after numbers on a web page

Key idea: the tests must pass on the ORIGINAL code in its ORIGINAL runtime (Python 2.7 in Docker) before any change is allowed.

Real-world test case

Michael Nielsen's "Neural Networks and Deep Learning" code

- Widely used, MIT-licensed, written for Python 2.6/2.7
- Author has said he will not port it to Python 3
- In scope: `mnist_loader.py`, `network.py`, `network2.py`
- Out of scope: `network3.py` (built on Theano, no longer maintained)
- Goal: run on Python 3.12 + current NumPy, with proof

Results

All numbers come from reports generated in the repository

23 → 0

Deprecated / Python-2-only usages

31 / 31

Tests pass on original code (Python 2.7)

0 / 29

Original code on Python 3.12: 0 pass, 29 fail

29 / 0

Modernized on Python 3.12: 29 pass, 0 fail

+ 2 intentional, documented behavior changes (integer-division fix)

Silent bugs caught

Problems a find-and-replace upgrade would miss

zip() iterator

In Python 3, zip() returns a one-time iterator. len() fails and data can only be read once.

Bare print

A bare print line silently does nothing in Python 3, so output quietly changes.

Integer division

eta/len(batch): $3/2 = 1$ in Python 2 but 1.5 in Python 3. Detected, documented, fixed on purpose.

How IBM Bob 2.0 powers LegacyLift

Every stage is a Bob task; every prompt is saved in prompts/

- Agent mode: multi-step tasks from analysis to verification
- Subagents: one per module, working in parallel
- Document understanding: official Python porting guide + NumPy docs
- Runs code: Python 2.7 and 3.12 test runners in Docker
- Built the report page and README from generated reports

Business value & what's next

Who it helps

Teams maintaining old internal tools, companies migrating off Python 2 and outdated libraries, and research groups keeping old code usable. Modernization becomes safe, reviewable and fast.

Next steps

- Java 8 → 17, COBOL paths
- CI integration
- Technical-debt dashboards

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